



GRADE 5

STEM OLYMPIAD

National Level Examination (NLE) 2026

SUBJECT CODE			
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Total Questions: 50

Total Marks: 50

Time: 75 Minutes

DO NOT OPEN THIS BOOKLET UNTIL INSTRUCTED TO DO SO

- Carefully Check Your Name, Class, Roll Number, And Other Required Details On The OMR Answer Sheet.
- Do Not Forget To Sign The OMR Sheet. Also Write Your Roll No. On The Question Paper Booklet.
- Use Only Black/Blue Ball Pen And HB Pencil To Fill The OMR Answer Sheet.
- All Questions Are Compulsory And Carry One Mark Each.
- Section A: Mathematics & Quantitative Reasoning (15 Questions); Section B: Science & Scientific Reasoning (15 Questions); Section C: Engineering, Technology & Data (10 Questions); Section D: STEM Olympiad Challenge (10 Questions).
- Only OMR Answer Sheets Will Be Evaluated.
- No Negative Marking For Incorrect Answers.
- You May Use The Space Provided On The Last Page For Rough Work (If Needed).
- Digital Devices Like Calculators, Smart Watches, Or Mobile Phones Are Not Allowed.
- After The Exam, Submit Both The OMR Sheet And The Question Paper To The Invigilator.

SECTION A – MATHEMATICS & QUANTITATIVE REASONING

Q1. A school had 48,706 notebooks. It distributed 19,858 notebooks to students and later received 6,975 more notebooks.

How many notebooks does the school have now?

- A. 34,823
- B. 35,823
- C. 36,823
- D. 37,823

Q2. What is 324×25 ?

- A. 7,100
- B. 8,000
- C. 8,100
- D. 8,200

Q3. What is 5,040 divided by 24?

- A. 200
- B. 210
- C. 220
- D. 240

Q4. Which fraction is greatest?

- A. $\frac{3}{5}$
- B. $\frac{5}{8}$
- C. $\frac{2}{3}$
- D. $\frac{7}{12}$

Q5. A ribbon is 8.5m long. Meera uses 2.75m and then another 1.8m. How much ribbon is left?

- A. 3.85m
- B. 3.95m
- C. 4.05m
- D. 4.15m

Q6. A rectangle has area 96 cm^2 and width 8 cm. What is its length?

- A. 10 cm
- B. 11 cm
- C. 12 cm
- D. 14 cm

Q7. A cube has edge 4 cm. What is its volume?

- A. 16 cm^3
- B. 32 cm^3
- C. 48 cm^3
- D. 64 cm^3

Q8. What is the smallest common multiple of 6 and 8?

- A. 12
- B. 18
- C. 24
- D. 48

Q9. What is the greatest common factor of 18 and 30?

- A. 3
- B. 6
- C. 9
- D. 12

Q10. A train leaves at 09:45 and travels 2 h 35 min. When does it arrive?

- A. 12:10
- B. 12:20
- C. 12:30
- D. 12:40

Q11. Which number does not belong to the pattern? 8, 16, 24, 35, 40

- A. 16
- B. 24
- C. 35
- D. 40

Q12. A bag has 5 red, 3 blue and 2 green balls. What fraction of the balls are blue?

- A. $\frac{1}{5}$
- B. $\frac{3}{10}$
- C. $\frac{1}{3}$
- D. $\frac{3}{5}$

Q13. The perimeter of a square is 44 cm. What is its side?

- A. 9 cm
- B. 10 cm
- C. 11 cm
- D. 12 cm

Q14. Which angle is 135° ?

- A. Acute
- B. Right
- C. Obtuse
- D. Straight

Q15. A bar graph shows 18, 25, 21 and 16 books read in four months. What is the average per month?

- A. 18
- B. 19
- C. 20
- D. 21

SECTION B – SCIENCE & SCIENTIFIC REASONING

Q16. Which sequence correctly shows the path of food through the digestive system?

- A. Mouth→stomach→food pipe→small intestine→Large intestine
- B. Mouth→foodpipe→stomach→small intestine→large intestine
- C. Mouth→small intestine→stomach→large intestine
- D. Mouth→stomach→large intestine→small intestine

Q17. Which condition is most important for a seed to germinate?

- A. Water, air and suitable temperature
- B. Only sunlight
- C. Only soil
- D. Only fertilizer

Q18. Which process in green leaves uses sunlight to make food?

- A. Respiration
- B. Photosynthesis
- C. Transpiration
- D. Germination

Q21. What happens to the shadow of a pole as the Sun changes position?

- A. Its length and direction can change
- B. It stays exactly the same
- C. It becomes a sound
- D. It turns into light

Q22. Which fuel is a fossil fuel?

- A. Coal
- B. Sunlight
- C. Wind
- D. Flowing water

Q23. Which action helps recharge groundwater?

- A. Covering all soil with concrete
- B. Allowing rainwater to soak into the ground
- C. Pumping water continuously
- D. Blocking ponds

Q24. Why is the small intestine important?

- A. Most digestion and absorption occur there
- B. It pumps blood
- C. It filters air
- D. It makes bones

Q25. Which adaptation helps a cactus conserve water?

- A. Broad soft leaves
- B. Leaves reduced to spines
- C. Webbed feet
- D. Gills

Q19. Which substance is most likely to dissolve completely in water?

- A. Sand
- B. Salt
- C. Chalk powder
- D. Sawdust

Q20. Which method is best to separate tea leaves from tea?

- A. Filtration
- B. Magnetism
- C. Evaporation
- D. Freezing

Q26. Which natural event is measured using a seismograph?

- A. Earthquake
- B. Rainfall
- C. Wind direction
- D. Temperature

Q27. Which simple machine is a seesaw?

- A. Lever
- B. Pulley
- C. Screw
- D. Wedge

Q28. A magnet attracts which material strongly?

- A. Iron
- B. Wood
- C. Plastic
- D. Glass

Q29. Why does a metal spoon feel hotter than a wooden spoon in hot soup?

- A. Metal conducts heat better
- B. Wood creates cold
- C. Metal is lighter
- D. Wood absorbs light

Q30. Which is a renewable source of energy?

- A. Natural gas
- B. Petroleum
- C. Solar energy
- D. Coal

SECTION C – ENGINEERING, TECHNOLOGY & DATA

Q31. A model bridge must carry more load. Which change is most useful?

- A. Add triangular bracing
 - B. Remove supports
 - C. Use longer unsupported spans
 - D. Wet the material
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Q32. A simple circuit has a cell, wires and bulb. The bulb glows only when:

- A. the circuit is closed
 - B. one wire is missing
 - C. both cell terminals are disconnected
 - D. the switch is open
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Q36. A student tests which soil holds the most water. What should be the same for all soil samples?

- A. Amount of soil and water added
 - B. Type of soil
 - C. Result recorded
 - D. Colour of label only
-

Q37. Which structure is strongest for a tall tower model?

- A. Broad base with cross-bracing
 - B. Very narrow base
 - C. Heavy top with no supports
 - D. One thin vertical stick
-

Q38. A rain gauge is used to measure:

Q33. A solar panel produces less electricity in shade because it receives less:

- A. light energy
- B. mass
- C. friction
- D. sound

Q34. A 2 kg load is lifted using a pulley. The main advantage of a fixed pulley is that it:

- A. changes the direction of force
- B. removes gravity
- C. makes the load massless
- D. creates energy

Q35. Which measurement best tells how fast a toy car moves?

- A. Distance travelled per unit time
- B. Its colour
- C. Its mass only
- D. Number of wheels only

- A. rainfall
- B. temperature
- C. wind speed
- D. air pressure

Q39. If a 500 mL bottle fills a tank 12 times, how much water enters the tank?

- A. 4 L
- B. 5 L
- C. 6 L
- D. 12 L

Q40. A robot starts facing North, moves 4, turns right and moves 6, turns right and moves 4. Where is it relative to the start?

- A. 6 steps East
- B. 6 steps West
- C. 4 steps North
- D. 4 steps South

SECTION D – STEM OLYMPIAD CHALLENGE

Q41. A plant is watered equally but one set receives 8 h light and another 2 h. Which variable is being changed?

- A. Amount of light
- B. Amount of water
- C. Plant type
- D. Soil mass

Q42. A water filter removes visible dirt but not all germs. What should be done before drinking if safety is uncertain?

- A. Use an appropriate purification method such as boiling
- B. Add more dirt
- C. Leave it open
- D. Mix it with soil

Q43. A 1.2 m board is cut into 4 equal pieces. Each piece is:

- A. 20 cm
- B. 30 cm
- C. 40 cm
- D. 60 cm

Q44. A device uses 3 cells of 1.5 V each in series. What is the total voltage?

- A. 1.5 V
- B. 3.0 V
- C. 4.5 V
- D. 6.0 V

Q45. Which design reduces friction in a moving cart?

- A. Use wheels with smooth axles
- B. Drag it on sandpaper
- C. Add rough pads to the axles
- D. Remove the wheels

Q46. A class records temperatures 24°C, 26°C, 28°C, 30°C at equal times. If the pattern continues, the next reading is:

- A. 31°C
- B. 32°C
- C. 33°C
- D. 34°C

Q47. A 20-tooth gear turns a 40-tooth gear. Compared with the first gear, the second gear turns:

- A. more slowly
- B. at the same speed
- C. twice as fast
- D. without moving

Q48. Which evidence best shows that air occupies space?

- A. An inverted glass traps air and stops water entering fully
- B. Air is invisible
- C. Wind can feel cool
- D. Clouds are white

Q49. A model house needs to stay cooler in summer. Which roof choice helps most?

- A. Light-coloured reflective roof with insulation
- B. Black metal sheet without insulation
- C. Transparent roof
- D. Dark roof under direct Sun

Q50. A seedling experiment gives heights 6, 9, 13, 18 cm on four days. The increases are +3, +4, +5. If the pattern continues, the next height is:

- A. 22 cm
- B. 23 cm
- C. 24 cm
- D. 25 cm

Sample paper